Mid-progress Report

XAI

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1. Introduction

This report documents progress on building a machine learning pipeline to predict diabetes risk using the BRFSS 2015 Health Indicators dataset. The focus of this XAI course project is not only predictive performance but also the interpretability and explainability of the model's decisions. A Random Forest classifier was implemented as the primary model and analysed using SHAP (SHapley Additive exPlanations) to explain individual and global feature contributions.

2. Dataset Overview

The BRFSS (Behavioral Risk Factor Surveillance System) 2015 dataset contains health survey responses collected across the United States. The binary target variable Diabetes_binary indicates diabetes diagnosis.

Key Characteristics

- Target: Diabetes binary (0 = No Diabetes, 1 = Diabetes)
- Class imbalance: 86% non-diabetic vs 14% diabetic
- 10 features selected covering cardiovascular, lifestyle, and demographic domains
- Train/test split: 80% - 20%, stratified to preserve class distribution

Selected Features

Feature	Type	Description
HighBP	Binary	High blood pressure diagnosis
GenHlth	Ordinal (1–5)	Self-rated general health
HighChol	Binary	High cholesterol diagnosis
BMI	Continuous	Body Mass Index
Age	Ordinal (1–13)	Age group scale
DiffWalk	Binary	Difficulty walking or climbing stairs
HeartDiseaseorAttack	Binary	History of heart disease or attack
PhysHlth	Continuous (0–30)	Days of poor physical health (past 30 days)
MentHlth	Continuous (0–30)	Days of poor mental health (past 30 days)
Income	Ordinal (1–8)	Income level category

3. Data Preprocessing Pipeline

3.1 Outlier Clipping

For continuous features with more than 15 unique values (BMI, Age, PhysHlth, MentHlth), outliers were capped using the IQR method. Values below Q1 - 1.5×IQR or above Q3 + 1.5×IQR were clipped to the boundary values, preventing extreme values from dominating tree splits.

3.2 Mean Imputation

A SimpleImputer with mean strategy handled missing values. The imputer was fitted exclusively on the training set and applied to both sets to prevent data leakage.

3.3 SMOTE Oversampling

The 86:14 class imbalance was addressed using Synthetic Minority Oversampling Technique (SMOTE), applied only on the training set after splitting. SMOTE generates synthetic samples for the minority class by interpolating between existing minority instances, ensuring no artificial patterns leak into the test evaluation.

4. Model: Random Forest Classifier

4.1 Model Description

A Random Forest ensemble was selected as the primary model. Random Forest builds multiple decorrelated decision trees and aggregates their predictions by majority vote. Key advantages relevant to this XAI project include: robustness to outliers and mixed feature types, native feature importance scores, and compatibility with SHAP TreeExplainer for efficient Shapley value computation. The following hyperparameters were used:

- n_estimators = 200 (number of decision trees)
- max_depth = 8 (limits depth to control overfitting)
- random_state = 42 (reproducibility)

4.2 Performance Results

Metric	Value	Interpretation
Accuracy	78.82%	Overall correct predictions
ROC AUC	0.8112	Good discriminative ability
Recall — Class 1	61.27%	Detects ~61% of diabetic cases
Precision — Class 1	35%	35% of flagged cases are truly diabetic
F1-Score — Class 1	0.45	Moderate balance on minority class

4.3 Result Analysis

<u>Class 0 (No Diabetes)</u>

The model performs strongly on the majority class: precision = 0.93, recall = 0.82, F1 = 0.87. This reflects the abundance of training examples for non-diabetic individuals.

<u>Class 1 (Diabetes)</u>

Performance on the minority class is weaker: precision = 0.35, recall = 0.61, F1 = 0.45. SMOTE significantly improved recall, but precision remains low, meaning many non-diabetic individuals are flagged as diabetic (false positives). For a clinical screening context, high recall is generally preferred over high precision — missing a diabetic case carries greater risk than a false alarm.

ROC AUC

A ROC AUC of 0.811 indicates good discriminative ability across all classification thresholds, and is a more informative metric than accuracy under class imbalance.

5. Explainability Analysis (SHAP)

5.1 Why SHAP?

Random Forest is an inherently black-box model while it provides aggregate feature importances, it does not explain individual predictions. SHAP (SHapley Additive exPlanations) addresses this by computing each feature's marginal contribution to every individual prediction, grounded in cooperative game theory. This makes SHAP a theoretically sound, model-agnostic explainability method that satisfies key XAI axioms: local accuracy, missingness, and consistency.

In this project, the TreeExplainer was used for computational efficiency with Random Forest, computing exact Shapley values by exploiting the tree structure.

5.2 Native Feature Importance vs. SHAP

The table below compares the two importance rankings. SHAP provides a more accurate picture of marginal contribution since native RF importance can be biased toward high-cardinality features.

Rank	Feature	RF Importance	SHAP |value|	Direction
1	HighBP	0.307	~0.11	High = Risk+
2	GenHlth	0.269	~0.10	High = Risk+
3	HighChol	0.184	~0.07	High = Risk+
4	BMI	0.066	~0.05	High = Risk+
5	Age	0.053	~0.04	High = Risk+
6	DiffWalk	0.071	~0.02	High = Risk+
7	HeartDiseaseorAttack	0.031	~0.01	High = Risk+
8–10	Income, MentHlth, PhysHlth	< 0.01	~0.01	Mixed

5.3 SHAP Plots

Two SHAP visualisations were generated for the positive class (Diabetes = 1):

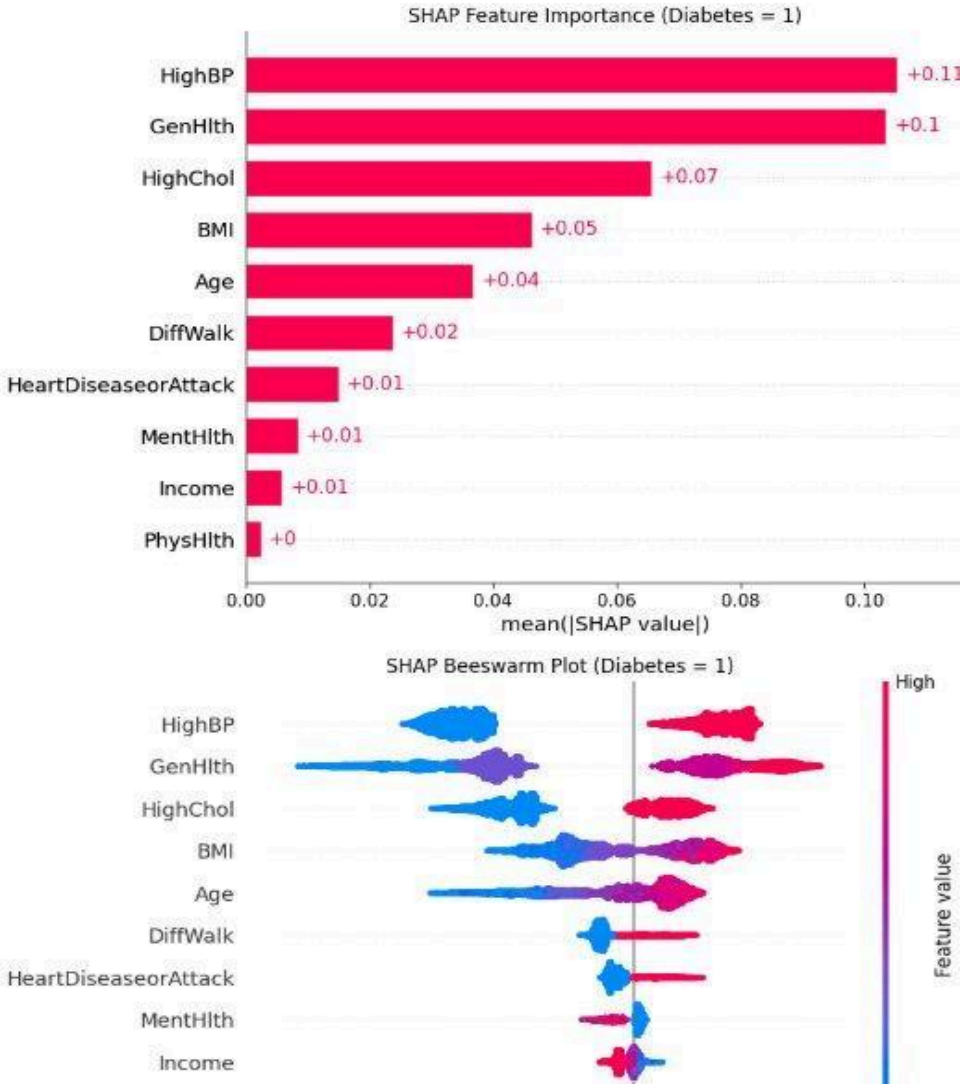

Figure : SHAP Feature Importance bar plot (top) and Beeswarm plot (bottom) for Diabetes = 1

Bar Plot (Global Importance)

The bar plot shows mean absolute SHAP values across all test samples. HighBP and GenHlth dominate with mean SHAP values of 0.11 and 0.10 respectively, followed by HighChol at 0.07. The remaining features contribute considerably less, with PhysHlth and MentHlth showing near-zero impact

suggesting they add minimal explanatory value for this model.

Beeswarm Plot (Global + Directional)

The beeswarm plot adds directionality to the global view. Each dot represents one test sample, positioned on the x-axis by its SHAP value and coloured by feature value (pink = high, blue = low):

- HighBP: Having high blood pressure (high feature value, pink dots) consistently pushes predictions toward diabetes

the largest and most consistent positive contributor.

- GenHlth: High values (poor health, coded 5) shift predictions strongly toward diabetes; low values (excellent health) are protective.
- HighChol: Mirrors HighBP's

pattern presence of high cholesterol increases predicted risk.

- BMI: A continuous gradient

higher BMI values produce higher positive SHAP contributions, consistent with obesity as a primary T2DM risk factor.

- Age: Older age groups push predictions toward diabetes, though with wider spread than the top three features.
- Income: An important socioeconomic signal

low income (low feature value, blue dots) increases predicted risk, consistent with health disparity literature.

5.4 XAI Insights Summary

The SHAP analysis reveals that the model's predictions are driven primarily by clinical comorbidities (HighBP, HighChol) and self-reported health status (GenHlth). This is clinically coherent and increases trust in the model's decisions. The beeswarm plot further confirms that the model captures non-trivial feature interactions

for instance, the spread in SHAP values for Age and BMI suggests heterogeneous effects across subgroups, warranting further individual-level explanation in future work.

6. Challenges Encountered

6.1 Class Imbalance

The 86:14 class imbalance was the most significant challenge. Without SMOTE, the model heavily favoured predicting class 0, producing near-zero recall on diabetic cases. SMOTE improved recall to 61% but introduced a precision-recall trade-off, reducing precision for class 1 to 35%. Future work should explore threshold tuning on the predicted probabilities to find a better operating point for the clinical use case.

6.2 SHAP Computation Cost

Computing SHAP values for a 200-tree Random Forest over ~50,000 test samples was computationally intensive. The TreeExplainer was used for efficiency, but runtime remained high. In practice, SHAP analysis was run on a representative subset of the test set. This may introduce minor approximation in global summary statistics.

6.3 Feature Relevance of PhysHlth and MentHlth

Both PhysHlth and MentHlth showed near-zero SHAP values and native importances below 0.006. While clinically plausible as indirect risk factors, they do not appear to contribute meaningfully to this model's predictions. Their inclusion may add noise and should be reconsidered in future iterations through formal feature selection.

6.4 Black-Box Nature of Random Forest

Despite SHAP providing post-hoc explainability, Random Forest remains a black-box model. SHAP explains what the model learned, not necessarily what is causally true. For an XAI course context, this distinction is important: high SHAP value for HighBP reflects the model's learned association, not a guaranteed causal relationship. Causal inference methods would be required to go beyond correlation.

7.Conclusion

The Random Forest classifier achieves a ROC AUC of 0.811 and a recall of 61% on diabetic cases, providing a solid predictive baseline. More importantly for this XAI course, the SHAP analysis yields interpretable and

clinically meaningful explanations: High Blood Pressure, General Health, and High Cholesterol are the strongest drivers of diabetes risk predictions, with directionality confirmed by the beeswarm plot.

The main limitation is the low precision for the diabetic class (35%), stemming from the class imbalance problem. The SHAP analysis also highlights that two features (PhysHlth, MentHlth) contribute negligible explanatory value. These findings will guide the next phase of model refinement and the implementation of a comparative second model.